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Leveraging Drilling Automation to Scale-Up Digital Drilling in Oil & Gas in One of the Industry's Largest Projects

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Abstract

The deployment of SLB's drilling automation system (DAS) in Saudi Arabia represents the largest implementation of autonomous drilling technology globally, revolutionizing digital drilling and enabling the transition from manual to digital operations. This paper will discuss the technical, operational and strategic aspects of the deployment, highlighting integration challenges and the impact on drilling efficiency and HSE. The objective is to provide insights into the complexities of large-scale drilling automation while offering a roadmap for future implementations in high-performance environments.

The drilling automation system runs on a local stack instead of a fully cloud-based system, ensuring compatibility with Saudi-specific requirements. Our intelligent planning platform generates the digital drilling program, which is uploaded to the local stack and then ingested into the DAS. This approach ensured the solution was customized to meet local requirements. The deployment followed a structured process, beginning with rig surveys to assess readiness and define modifications. Hardware installation and system configuration were completed, followed by commissioning through rigorous testing and troubleshooting. The final phase involved on-site hyper-care to ensure stable operation and optimize automation execution.

The deployment of the DAS in Saudi Arabia spanned 11 deep-drilling rigs (2,500 HP) designed for gas production, handling complex well profiles including ERD wells. The system automated key operations such as pre-connection, on-bottom drilling and post-connection sequences significantly improving operational efficiency. One major challenge was integrating the DAS across three distinct rig-control systems (NOV, Cameron and HH), each requiring unique adaptations. Frequent system updates added complexity, demanding real-time troubleshooting and continuous refinement. Another key challenge was shifting the client's mindset from conventional automation to multi-workflow autonomy, requiring extensive education on performance metrics and system benefits. The project also faced pressure to deliver rapid results despite being in an early stage, necessitating a balance between short-term wins and long-term optimization. Ultimately, the DAS demonstrated its potential to transform drilling operations, reducing human intervention, improving HSE conditions and optimizing well delivery. The insights gained from this deployment highlight the best practices for large-scale automation rollouts, emphasizing the importance

of robust integration strategies, stakeholder alignment and iterative system enhancements to maximize automation benefits.

This paper presents the first large-scale deployment of a Multi-Workflow Autonomy DAS in a high-performance environment. It offers new insights into overcoming integration complexities, managing stakeholder expectations and optimizing automation efficiency in real-time drilling operations. By sharing lessons learned, best practices and future development pathways, this study provides valuable guidance for industry professionals seeking to implement and scale autonomous drilling solutions.

Introduction

In one of the largest digital drilling initiatives globally, a major gas well delivery project in Saudi Arabia became the proving ground for the first large-scale deployment of a multi-workflow autonomous drilling solution. While prior optimization efforts had already brought manual drilling performance close to its technical limit, achieving further improvements required a disruptive shift, one that could redefine the operational model altogether. To drive the next step-change in performance, drilling automation was identified as the key enabler.

SLB's proprietary autonomous drilling system developed since 2014, was selected for deployment due to its ability to control surface equipment, execute drilling tasks autonomously and adapt in real time to subsurface conditions. Designed to eliminate variability introduced by human operation, the system executes predefined workflows precisely while leveraging intelligent engines to optimize parameters and prevent dysfunctions.

The initial deployment in the gas project in 2020 was done on 2 rigs out of the 11 available rigs as a pilot. The findings of this pilot deployment, detailed in paper SPE-203251-MS, highlighted a clear improvement in various drilling performance metrics mainly a 19% increase in ROP, an 11% increase in downlinking efficiency and a 23% decrease in connection time with the most active contributor to these results being the increase in operational consistency. This successful pilot project paved the way to expand from 2 rigs which both used the same Rig-Control-System to all 11 drilling rigs which use 3 different RCSs each having their own specific deployment method and challenges.

The DAS empowers drilling operations with advanced AI-driven workflows, ensuring optimal performance and significantly reducing manual interventions. Designed to enhance both decision-making and execution, this solution integrates real-time recommendations and fully autonomous processes to deliver superior results in well delivery. Key features include:

- **AI-Driven Automation:**
 - AutoROP: Automatically adjusts drilling parameters to optimize performance
 - AutoSteering: Maintains trajectory accuracy with minimal manual input
 - Goal-Based Autonomy: Allows teams to set performance objectives while the system autonomously executes operations
- **Operational Integration:**

Seamlessly integrates with procedural adherence and reporting systems, ensuring comprehensive oversight of drilling activities.

This deployment has had transformative effects on LSTK projects, delivering measurable improvements in efficiency, consistency and cost management:
- **Performance Optimization:**
 - Achieved improvement in on-bottom ROP, reducing project timelines and improving productivity

- Enhanced wellbore trajectory control, minimizing rework and ensuring precision in high-value projects
- Reduced Manual Effort:
 - Significantly reduced reliance on human intervention, leading to fewer errors and improved operational consistency
- Streamlined Execution:
 - Autonomous workflows ensured seamless execution of complex drilling processes, minimizing disruptions and delays.

SLB's DAS represents a pivotal shift in how LSTK projects are managed and executed. By leveraging AI to optimize performance and reduce manual interventions, this solution has elevated the standard for well delivery. The leadership in integrating and scaling these technologies has ensured that they deliver maximum value and transforming the operational landscape of turnkey contracts.

This paper focuses on the deployment across these 11 deep gas rigs. It outlines the strategic, technical and operational frameworks that supported the implementation and shares key lessons learned while navigating challenges such as integration across multiple rig-control systems, frequent system updates and the cultural shift required to adopt multi-workflow autonomy. By documenting the roadmap, outcomes and insights gained, this study aims to guide future large-scale autonomous drilling projects in similarly demanding environments.

Terminology

Given the diversity of disciplines and global regions within the oil and gas industry, terminology, particularly in the realm of digital technologies, can vary significantly. As new automation solutions emerge and evolve, a consistent understanding of key terms becomes essential for clear communication. The following definitions are provided to establish a common reference framework within the context of this paper:

- I. **Drilling Automation System (DAS):** A software platform designed to autonomously operate rig surface equipment with minimal or no human intervention.
- II. **Rig Control System (RCS):** The centralized control unit responsible for managing, monitoring, and executing commands to rig floor systems, typically operated by the driller through a human-machine interface (HMI).
- III. **Automation:** The application of digital technologies to perform operational tasks with limited human input. This is distinct from mechanization, which refers to the use of equipment or machinery that remains under direct human control to perform physical functions. For the purposes of this paper, Automation can be described as the utilization of an Auto-Driller functionality for holding inclination and azimuth with on-bottom drilling.
- IV. **Single-Workflow Autonomy:** This presents a higher level of autonomy above drilling Automation. Single-Workflow Autonomy describes the system's ability to perform a workflow completely independently of human intervention. A workflow can be described as a series of operations which are done sequentially to complete a certain action (e.g. the "Pre-Connection" workflow contains all the operations taken after drilling a stand in preparation to set the stand in slips to perform a connection).
- V. **Multi-Workflow Autonomy:** This presents the final level of autonomy prior to "Full Autonomy". Multi-Workflow Autonomy describes the system's ability to conduct multiple workflows independently without interruption and without the need for a human input to signify the switch from one workflow to the next (e.g. the DAS's ability to go from "Post-Connection" workflow to "Drill-a-Stand" workflow to "Pre-Connection" workflow without any human intervention).

- VI. **Full Autonomy:** This presents the final level of drilling automation. Full Autonomy describes a system which can handle the entire drilling process from start to finish with no need for any human input.
- VII. **CODA / HPE:** The central computing unit that hosts the DAS.
- VIII. **Interactive Driller Display (IDD):** The IDD is the driller's primary interface with the DAS.
- IX. **Digital Drilling Program (DDP):** The intelligent planning platform generates the DDP to be then ingested by the DAS to enable drilling automation.
- X. **Original Equipment Manufacturer (OEM):** The company that makes components or products used in another company's final product.

The Automation Journey

The journey of deploying the DAS across operations marked a transformative phase in how drilling programs were planned, executed and continuously optimized. This evolution demanded changes not only in workflows and tools, but also in the mindset of the organization from engineers and drillers to project managers and rig contractors.

One of the earliest shifts observed was the transition from traditional manual planning methods to a more data-driven and iterative planning model. Prior to the deployment of the DAS, the drilling program generated by the intelligent planning platform was a static document prepared in advance and rarely modified once operations began with no utilization of the Digital Drilling Program (DDP). However, with the deployment of SLB's Drilling Automation System (DAS), the DDP was utilized to feed into the digital workflows that drive drilling automation. Automation significantly increased the value for collaboration between engineering, operations and digital teams. The successful design and deployment of automated workflows required domain experts from different functions to come together. Engineers needed to translate their drilling knowledge into machine-readable instructions, while digital teams worked to embed these into the software. Operations teams, in turn, provided on-the-ground feedback to ensure the practicality of the proposed solutions. This tight alignment across functions was essential to maximize the benefits of automation.

The automation journey also brought about a redefinition of roles as engineers started taking on more real-time operational involvement while rig crews needed to understand digital workflows. This required significant upskilling for drillers who had to become proficient in managing automated systems, engineers who needed to understand the constraints and conditions at the rig site and cross-functional teams who had to operate with a shared language and set of objectives.

Deploying the DAS across rigs from different RCSs presented a host of technical and operational challenges. Each rig design had its own set of control systems, hardware interfaces and operating procedures, requiring the automation system to be both modular and highly configurable. This complexity necessitated a flexible architecture for the DAS, allowing the system to adapt to rig-specific conditions while maintaining consistency in performance. Rig contractors played a vital role in enabling this flexibility. Close collaboration with contractors ensured that integration between the DAS and the rig control systems was successful. These partnerships allowed for smoother field trials, effective troubleshooting and faster deployment cycles. The early involvement of rig contractors in the planning and testing phases significantly reduced the risk of deployment delays and system incompatibilities.

New SOPs had to be developed to govern the operation of automated workflows, including parameters for initiation, execution and manual intervention. These procedures were designed to balance the reliability of automation with the situational awareness of experienced personnel. Simulation and testing were critical tools in this development. Before field deployment, teams used digital twins and simulators to validate workflow behavior, adjust boundary conditions and refine exception handling logic. This process minimized disruptions during live operations and built user confidence in the system.

Recognizing that successful deployment hinged on user competency, a comprehensive training program was introduced using supervised hands-on field exposure. The goal was not only to familiarize personnel with the system's functionality but to empower them to intervene, adapt workflows and troubleshoot issues as needed. Automation also led to revisions in safety protocols. Teams needed to account for potential system failures, unexpected behavior or misconfiguration. Emergency stop procedures, system overrides and fallback modes were clearly defined and incorporated into the SOPs. Rig crews were trained on these updated protocols to ensure they could act swiftly and safely in any scenario.

The success of this DAS deployment was evaluated through multiple KPIs, including:

- Reduction in drilling time per section
- Cost savings due to improved efficiency
- Reduction in downhole incidents
- Improved consistency across wells and rigs
- Increased crew confidence and reduced human error

These KPIs allowed the team to quantify the value added by automation and identify areas that needed further improvement. Feedback from the field was a cornerstone of the deployment journey. Drillers, engineers and automation support personnel regularly provided input on what worked, what didn't and how workflows could be improved. This feedback was reviewed alongside operational data using analytics dashboards and root-cause analysis techniques. As a result, workflows were continuously refined, safety margins adjusted and new features added.

Each deployment provided new insights and these were systematically captured and shared across teams. Learnings from early deployments were instrumental in shortening the timeline for subsequent rollouts. Teams created playbooks, checklists and onboarding kits based on these lessons to help standardize future deployments. Importantly, this knowledge-sharing culture helped embed continuous learning into the automation journey, ensuring that both technology and people evolved together.

How SLB's DAS Works Using AutoRop Engine

The drilling automation system (DAS) deployed in this study was originally developed as the execution engine of a broader well construction platform, as detailed in A. J. (2020). This platform was conceived to bridge the gap between planning and execution by enabling multi-stakeholder collaboration during the well design phase and translating that plan into a machine-readable digital drilling program. Once validated by rig personnel, DAS could autonomously execute predefined drilling actions while continuously capturing operational data to inform future planning cycles.

The ROP Optimization Engine leverages the drilling roadmap generated by the intelligent planning platform. This roadmap incorporates data from offset wells, technical limits, and industry best practices, serving as a set of "guardrails" to establish a safe operating envelope for drilling. Within this envelope, the engine computes optimal drilling parameters to maximize performance while maintaining safety. At the core of the ROP Optimization Engine is a modeling approach based on the DeTournay principle, which quantifies the relationships between key drilling parameters. For example, it defines how torque responds when a unit of weight on bit (WOB) is applied. By integrating this principle with the safe operating envelope derived from the roadmap, the engine can interpolate recommendations and predict whether any operational constraints might be exceeded at any given point.

The output of the ROP Optimization Engine includes recommended values for controllable parameters such as rotary speed (RPM) and weight on bit (WOB). Additionally, it provides predictions for associated metrics, including torque, differential pressure (in cases where a mud motor is used), and the expected

rate of penetration (ROP). A key feature of the system is its ability to update the formation model in real time. The engine continuously monitors drilling conditions on a second-by-second basis to evaluate whether the model's predictions align with the actual observed conditions. If discrepancies arise, indicating that the formation model is no longer accurate, the engine dynamically refreshes the model within seconds. This ensures that optimal drilling parameters are consistently applied throughout the operation, maximizing efficiency and performance while adapting to changing conditions.

By identifying the optimal drilling parameters that remain within the constraints defined by the drilling roadmap, the ROP Optimization Engine plays a critical role in preventing erratic behavior or dysfunctions commonly encountered when setpoints or control gains are set too high. Such dysfunctions in AutoDriller systems have become a significant contributor to bottom-hole assembly (BHA) failures and inefficient drilling performance, particularly in interbedded and high-temperature formations. Patusek et al. (2023) highlight that poorly tuned AutoDriller control systems can destabilize surface parameters such as weight on bit (WOB), rate of penetration (ROP), and differential pressure. This instability often results in critical issues, including bit damage, motor stalls, and stick-slip vibrations. Their study documented how targeted tuning of control loops, combined with specific drilling procedures, such as switching to ROP control mode during transitions between hard and soft formations, significantly reduced BHA trips and motor failures in wells across Oklahoma and East Texas. Notably, consistent ROP control maintained a stable depth of cut, minimizing dysfunction-induced damage to BHA components.

The ROP Optimization Engine addresses these challenges by determining the optimal ROP and WOB setpoints to effectively control the AutoDriller. This automation eliminates the need for frequent manual intervention by the driller, particularly in complex formations such as interbedded lithologies. As the ROP Optimization Engine computes the expected ROP, it uses this value as the ROP setpoint, which governs a constant drawworks speed and maintains a consistent depth of cut at the bit. By ensuring stable operational parameters, the engine promotes smoother drilling operations, mitigates dysfunctions, and extends the lifespan of critical drilling components.

During the pilot phase of this deployment, the drilling automation system (DAS) was initially implemented as a standalone solution, without full integration into the planning environment. However, following the success of the pilot, this integration was successfully achieved during the broader rollout. The planning environment, originally designed as a cloud-based platform, was adapted to operate on a localized stack to comply with in-country data residency requirements. This adaptation demonstrated the resilience and flexibility of the solution, enabling it to function effectively despite infrastructure limitations. The DAS was also deployed on the same local stack, which facilitated seamless ingestion of the "Digital Drilling Plan" from the planning environment and enabled end-to-end digital execution. The foundational technical architecture, operating principles and general functionality of the DAS are detailed extensively in [B. Goodkey \(2020\)](#), with a summarized overview provided in the following sections.

- I. **Dynamic Planning:** The system featured a real-time planning engine capable of adjusting to changes in the drilling environment. By continuously analyzing sensor inputs, it could detect deviations, reassess the current operational state, and re-sequence activities to meet updated objectives. This allowed for seamless transitions between manual and automated modes, including the autonomous handling of unplanned events such as formation changes or drilling dysfunctions.
- II. **Safety & Resilience:** The automation system was intentionally designed as a non-critical overlay to the existing rig control system (RCS). In the event of failure or loss of communication, full control would default back to the driller, ensuring uninterrupted well delivery. All system commands passed through the RCS gateway, maintaining standard operational safeguards and allowing the driller to override or revoke automation at any time.
- III. **Interoperability:** To accommodate the wide range of rig control systems used across the industry, DAS was developed as a hardware-agnostic solution. A single, customized interface linked the

automation software to the specific RCS in use, allowing for scalable deployment across rigs from different manufacturers. The interface could be disengaged at any time, preserving the integrity and operability of the original rig systems.

This modular and resilient design enabled the DAS to be effectively deployed across multiple high-spec rigs in KSA, establishing a foundation for autonomous operations in complex drilling environments. The drilling automation system (DAS) deployed in this study operates through a set of interlinked functional modules working in parallel to achieve the selected drilling objectives. As outlined in R. Gallo (2017), the broader system architecture includes capabilities such as autonomous ROP optimization, automatic downlinking, survey handling, steering, tripping, and wellbore monitoring.

A core enabler of the DAS's adaptability is its AI Planner framework, which underpins the system's ability to manage complex operations with minimal manual intervention. The AI Planner interprets the current operational state and plans the most efficient path to the goal while respecting all domain boundaries by utilizing multiple domain engines to create a clear picture of downhole conditions to make informed, accurate decisions. This allows for seamless transitions between manual and automated control at any stage of the operation, without being restricted to predefined start points. This flexibility ensures that automation can be engaged or disengaged dynamically, depending on real-time rig conditions and operational context. The system supports on-demand re-planning, allowing it to rapidly adjust sequences when user priorities shift or unexpected drilling conditions arise. Its situational awareness enables the execution of remedial actions that are fully responsive to live sensor data, ensuring that all decisions remain grounded in current operational realities. Perhaps most critically, the AI Planner (as described in Figure. 1) provides a scalable foundation for increased levels of autonomy. New workflows can be introduced by modifying planning rules, eliminating the need for full system redesigns. This modularity not only accelerates the evolution of autonomous capabilities but also enhances the system's ability to adapt to future operational needs.

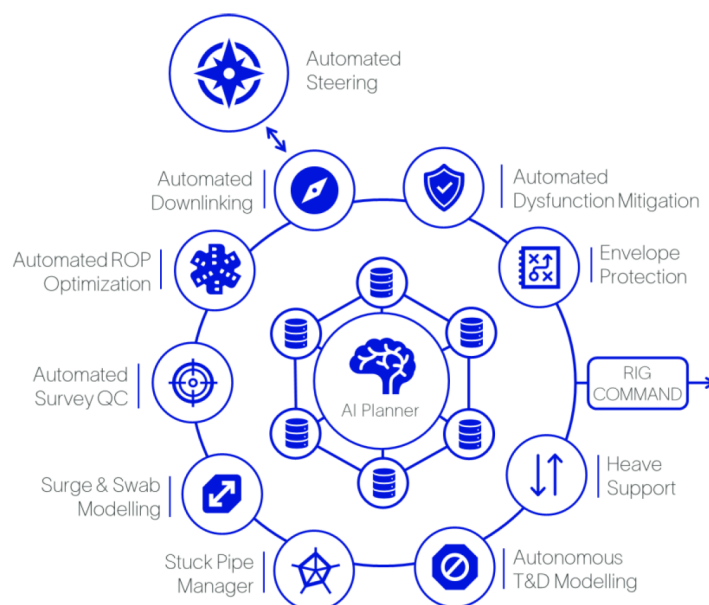


Figure 1—AI Planner framework highlighting what input data are needed by the automation system to be then translated as "Rig Command".

During the pilot phase described in this study, only four of the previously mentioned modules were deployed in operation (Drill-a-Stand, ROP Optimization, Drilling Dysfunction Mitigation, and Downhole Tool Communication). Following the successful validation of the system, the full suite of automation

modules has since been implemented across the broader deployment. A description of each of these modules is included below.

Drill-a-Stand

This module autonomously executes all activities related to pre-connection, post-connection, and on-bottom drilling. By leveraging the system's dynamic planning engine, it interprets rig sensor data to identify the current state and sequences the required steps to reach the designated drilling goal. Unlike manual operations that are susceptible to human variability, DAS ensures procedural consistency and compliance with operator guidelines, thereby reducing performance fluctuations and mitigating risks such as tool failure or stuck pipe. The complete Drill-a-Stand sequence can be seen in Figure 2.

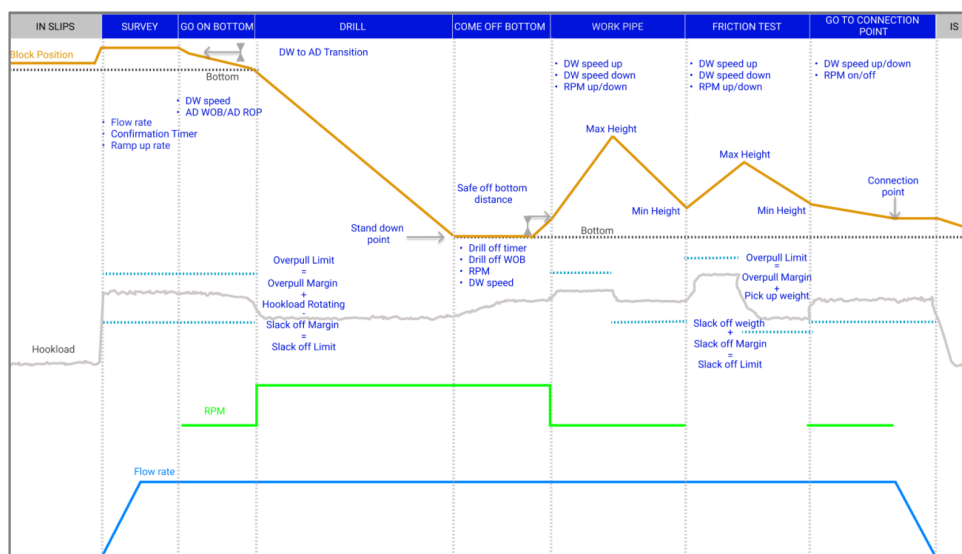


Figure 2—Drill-a-Stand sequence starts from the "In-Slips" status then proceeds to post-connection operations followed by the drilling of a stand then, finally, how the system performs pre-connection practices before releasing control to the driller to set the slips.

ROP Optimization

The ROP optimization engine continuously models drilling conditions in real time and adjusts drilling parameters to maximize efficiency. Unlike conventional auto drillers that optimize based solely on surface parameters, this engine incorporates formation behavior and depth-specific conditions. The result is a smarter, more responsive drilling process that adapts to changing downhole environments. The DAS consumes the same channels as every other industry solution and, because it takes time for parameter adjustments on surface to be felt downhole, the DAS uses a series of proprietary engines to convert the surface parameters into downhole parameters. This prevents from overapplying on surface while waiting on the response, then putting on full brakes once the parameters are felt downhole. From a few data points, the DAS builds a model of the formation based on these calculated downhole parameters.

This means that, from a few datapoints at a single parameter combination, it can understand the resulting ROP will be achieved at any combination of RPM and WOB without ever testing them. It will choose the most efficient combination that achieves the best performance (meaning that hole cleaning and efficiency is prioritized; the DAS may select more conservative parameters that achieve the same performance). Then, a second AI algorithm is operating in parallel to the formation modelling acts as a third-party consultant which audits the health of the formation model that is active. If at any point the second algorithm detects the formation model is no longer accurate (if formation characteristics have changed), it will force the system to build a new model based on the current formation (similar to running machine learning on-the-fly on the current formation data).

This means that adapting to new formations takes seconds. For AutoROP to work, it needs the input surface and downhole data and the limits for various parameters. Then, at every second, AutoROP creates a model with the WOB, torque and the Depth of cut per revolution and plotting this chart in 3 dimensions as seen in Figure. 3. This creates a model for each formation, represented in graph x with different colors for each model. Given that the model can be extrapolated with the updated downhole data, the AutoROP engine can accurately determine the relationship between WOB, RPM, ROP and Torque.

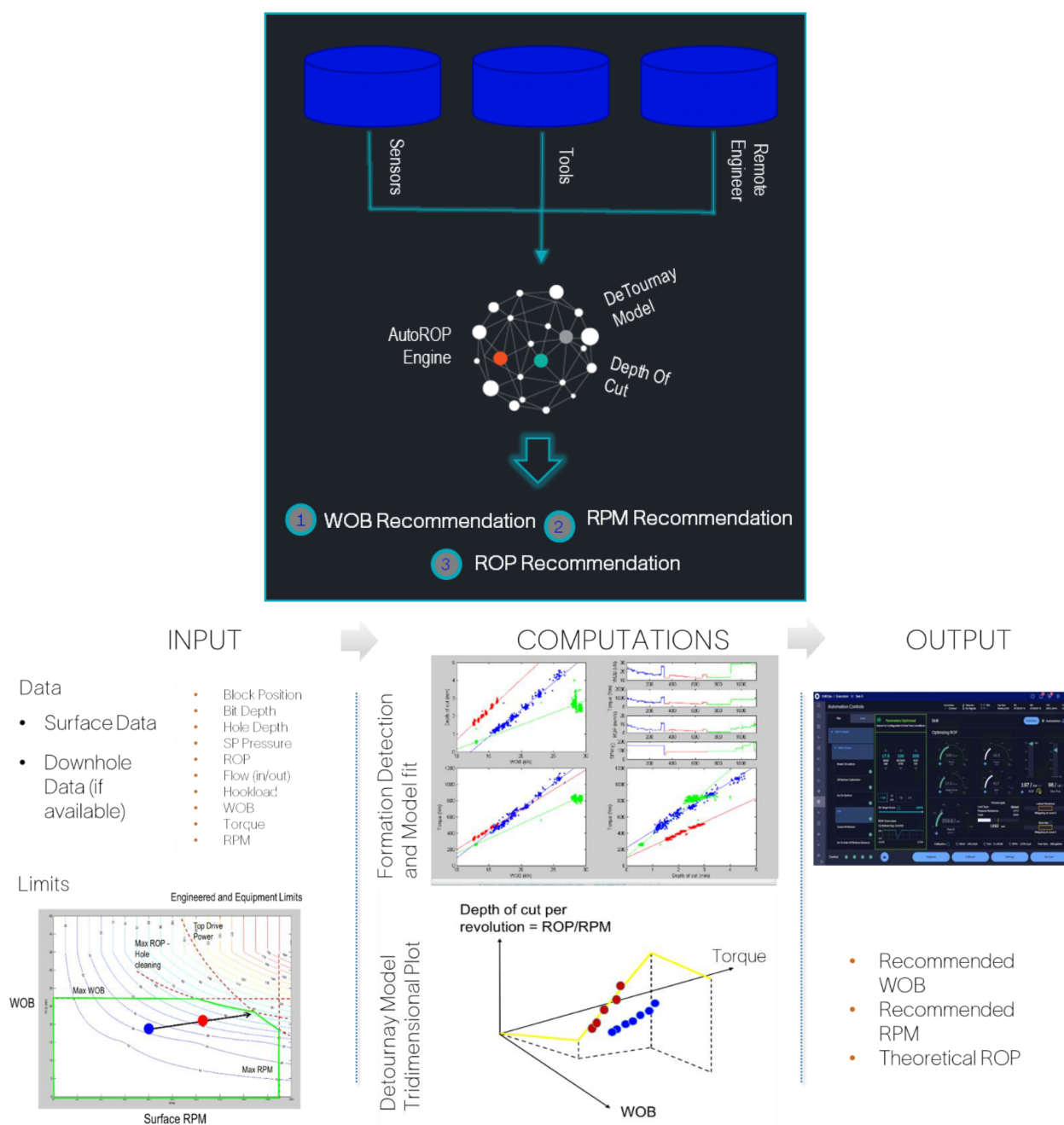


Figure 3—AutoROP engine uses drilling parameters and downhole conditions to create a model with the WOB, torque and the Depth of cut per revolution in 3 dimensions.

Drilling Dysfunction Mitigation

To address common issues such as stick-slip, shock, and vibrations, the DAS monitors surface and downhole data to detect dysfunctions and apply appropriate mitigation strategies. The system executes these actions

autonomously, escalating intervention levels based on the severity and persistence of the event. This automated response not only reduces tool failure risk but also ensures compliance with mitigation protocols that are often inconsistently applied in manual operations.

As illustrated in Figure 4, S&S Mitigation Example, the system follows a structured sequence of actions when addressing drilling dysfunctions through the ROP Optimization module. The process begins with (1) detection, triggered when stick-slip severity exceeds the predefined threshold specified in the system configuration. In response, the ROP Optimization engine initiates mitigation by reducing drilling parameters, typically Weight-on-Bit, to alleviate torque levels. If the dysfunction persists, the system escalates to a second mitigation level (2) a stepwise reduction of Weight-on-Bit, with each setpoint held for a user-defined duration to assess its effectiveness. Once the stick-slip severity falls below the allowable limit, the system proceeds to (3) conclude the mitigation phase and gradually resumes ROP optimization. At this stage, (4) Weight-on-Bit is incrementally increased until the optimal setpoint is restored. If the condition remains unresolved, the system advances to a third and final mitigation stage, whereby it temporarily picks up off bottom to dissipate stored energy in the drill string before re-engaging with revised parameters. This mitigation strategy operates independently of, and remains fully compatible with, existing stiffness-damping systems such as conventional "soft-torque" configurations.

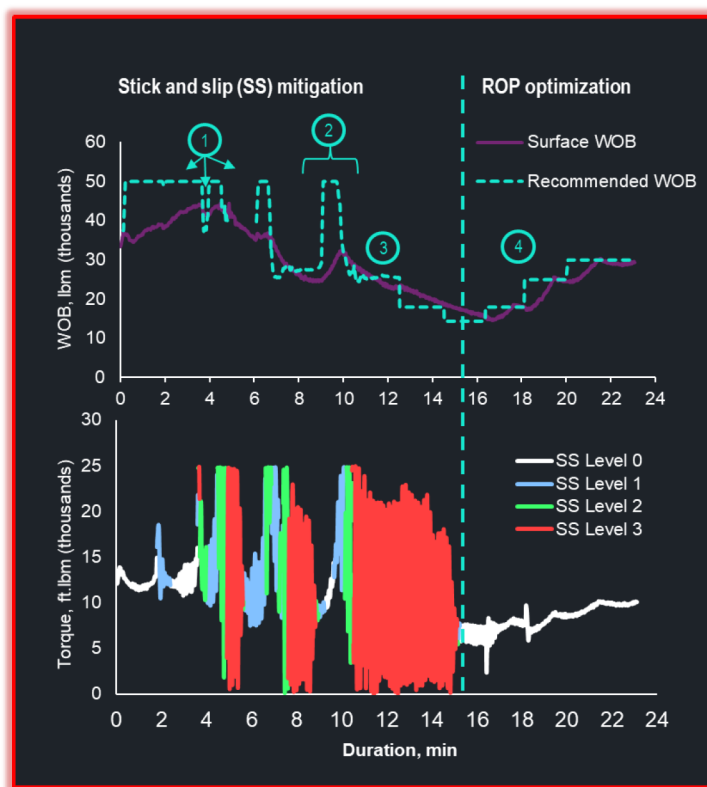


Figure 4—S&S mitigation sequence is explained versus surface WOB & time.

Downhole Tool Communication (Directional Automation)

Directional tools commonly require surface-downhole communication via flow or RPM-based telemetry downlinks. In manual settings, variability in execution often leads to failed or repeated commands, causing invisible lost time. The DAS automates the downlink process with precise control overflow and RPM patterns, significantly improving communication success rates and reducing non-productive time associated with retries or off-bottom downlinking.

Remote Operations & Integration with Other Drilling Environments

As drilling operations become increasingly complex, coupled with high crew turnover, remote operational capabilities have become essential for sustaining performance and operational continuity. A key advantage of the DAS lies in its integration within the broader SLB ecosystem. This integration enables seamless interaction across planning, execution, and post-well review workflows, facilitating collaboration among stakeholders throughout the well construction process.

The system supports remote monitoring and control capabilities, allowing town-based personnel to access real-time domain data, monitor progress, and submit change requests directly to the rig site. Executed automation results are automatically recorded and stored within the SLB technical suite of drilling-related environments, ensuring alignment with planning data and enabling data-driven post-job analysis. In addition, automation configuration, such as roadmaps and task sequences, can be retrieved from various other platforms, ensuring that operations follow the approved digital drilling program. This interconnected workflow not only enhances transparency and decision-making speed but also establishes a scalable model for collaborative drilling operations in a high-performance environment.

Envelope Protection

Human error continues to be a leading contributor to non-productive time (NPT) in drilling operations worldwide. To mitigate this risk, the DAS incorporates advanced envelope protection capabilities designed to proactively safeguard against critical operational failures such as stuck pipe and formation fracture.

The system leverages industry-leading detection algorithms developed to function autonomously, even in the absence of cloud connectivity or direct involvement from domain experts. A fully automated torque-and-drag (T&D) model continuously monitors operational trends and adjusts actions in response to emerging risks. Key protection mechanisms, such as overpressure and pack-off prevention, remain actively engaged throughout the operation to ensure real-time response to potentially hazardous conditions. Risk awareness is further enhanced through real-time visualization and event recording, allowing drillers to remain informed of both current threats and system-initiated mitigations. Additionally, the automation roadmap can be tailored to align with varying levels of operational risk, enabling dynamic segmentation and execution strategies based on the desired balance between performance and safety.

Together, these modules form the operational backbone of the deployed system, demonstrating how targeted automation capabilities can enhance drilling consistency, efficiency, and procedural compliance in complex well environments.

Deployment Process and Challenges

Given the tight operational tolerances in this project, successful deployment of the DAS required meticulous planning and thorough preparation. While each deployment is customized to fit the specifics of the target rig, the overall process follows a consistent sequence involving four main stages:

1. Planning
2. Execution
3. Monitoring
4. Closure

Planning

The deployment journey begins with a rig survey, risk assessment, stakeholders' engagement, logistic, procurement and deployment schedule. The rig survey determines rig's compatibility with the DAS. This step assesses whether the rig's infrastructure is a DAS candidate and identifies any modifications required to enable the deployment. Recommendations are categorized into mandatory changes (essential for drilling

automation functionality) and optional enhancements (to improve performance or reliability). The rig survey typically investigates:

- Rig electrical system whether it's AC VFD or DC SCR
- Rig Control System (RCS) manufacturer and interface development status with the DAS
- Rig equipment, mud pumps, drawworks, top drive, auto driller models and specifications.
- RCS integrated status for the mud pumps, drawworks, top drive, and auto driller
- Software versions of key systems (drawworks, top drive, auto driller)
- The optimum system hardware and network layout for the rig deployment
- Suitable power, internet and data source for the automation system

Risk assessment for automation system deployment is crucial because the system receives real-time data, runs computation and sends feedback to RCS to move rig equipment in response to downhole tools and formation changes. Therefore, any misconfiguration or oversight during deployment can lead to unsafe operations, equipment damage or even well control incidents. Furthermore, the automation system changes how engineers and rig crews interact with operating systems. A poor deployment without proper training or interface validation can lead to misuse or misunderstanding which would increase operational risk. In addition, any software bugs, network issues or sensor failures can compromise real-time analytics, predictive models or automated workflows. Having a very limited rig move and flat time on those eleven land rigs makes a very narrow window for the deployment time, so any failure during deployment can lead to non-productive time (NPT), increased cost and loss of stakeholder confidence.

Early engagement with the four key stakeholders (product owners, rig OEMs, rig contractors and the customer) played a pivotal role in the successful deployment of the project. Over thirty hours of combined virtual and in-person meetings are normally conducted to align expectations and define clear project objectives. These sessions enabled the team to proactively identify potential deployment risks and establish mitigation strategies, culminating in a well-defined execution plan. Coordination with logistics teams across all parties ensured smooth mobilization and timely importation of critical hardware into the country ahead of job execution. This included essential components which need to be sourced from multiple continents, properly stored in workshops and subjected to thorough physical inspections and software installations. The hardware list might vary from rig to rig based on rig survey, but it generally includes:

- CODA/ HPE units
- Integrated Driller Display (IDD)
- Ethernet switches
- Cables, connectors, and sockets
- Hand tools and mounting accessories for installing CODA/HPE and IDD

A full description of these components is provided in the following chapter.

Deployment of the drilling automation system in some rigs required up to 3 days of rig standby time. However, the eleven land rigs are actively drilling with very limited flat time between sections. They are moving fast from well to another, which creates a challenge for the team to develop an optimized deployment schedule with contingency plans in order to avoid operation interruption and non-productive time.

Execution

After completing the detailed planning phase for each rig deployment, the project advanced into the execution phase which is recognized as the most critical and complex stage of the entire process. This phase is divided into two key components: workshop preparation and in rig-site implementation.

Execution begins with software installation, network and firewall configuration, fabrication of rig-specific mounting accessories and assembling the necessary toolkit for system Hardware installation. Once workshop preparation is completed and before shipping equipment to the field, pre-deployment preparations are carried out by an expert Digital Drilling Engineer at the base location to minimize rig-time configuration. These include:

- Selecting the correct software version and referencing internal deployment notes
- Installing the appropriate operating system and commissioning software for both CODA and IDD
- Applying any necessary patches for CODA
- Creating a customized deployment layout for each rig
- Pre-configuring the network switch and firewall settings according to the rig layout

This phase significantly streamlines the field installation and reduces troubleshooting time. Upon successful completion of the workshop phase, the hardware is carefully packed and prepared for transport to the rig site.

Once System Hardware arrives at the rig site, the Deployment team will coordinate with drilling crew offloading hardware on the rig site. Then, the digital drilling engineer presents the installation and commissioning process to the rig crew and rig management team. After that, an internal safety meeting with the deployment team and an external safety meeting with the rig crew and the management team takes place before starting the job. During this meeting, DAS team coordinates with the rig crew and drilling contractor to organize the physical installation. This coordination ensures minimal disruption to ongoing operations and clearly defines task ownership. During discussions, the drilling automation team agrees with the rig crew, RCS technicians and rig management team on the following:

- The timeline for installation and go-live
- The role of the RCS provider (e.g., HH, NOV) and their scope of work
- Communication protocols between the rig and DAS teams
- Detailed documentation and drawings provided to the drilling contractor outlining installation requirements

The final phase involves validating the integration between the DAS and the rig's control systems. This step ensures all components are fully functional before live operations begin. The outcome is a jointly signed I&C document endorsed by both the RCS OEM and the DAS team. The I&C process includes:

- RCS software upgrade, enabling it to interpret and execute DAS commands
- RCS functionality testing, conducted according to standardized commissioning procedures
- DAS testing, ensuring it operates correctly and communicates seamlessly with RCS

System installation and commissioning have now been completed, comprising the RCS interface with the CODA, IDD for the driller, client workstation access and the DAS data acquisition system. A sample deployment layout is shown in the following image (refer to [Figure 5](#)).

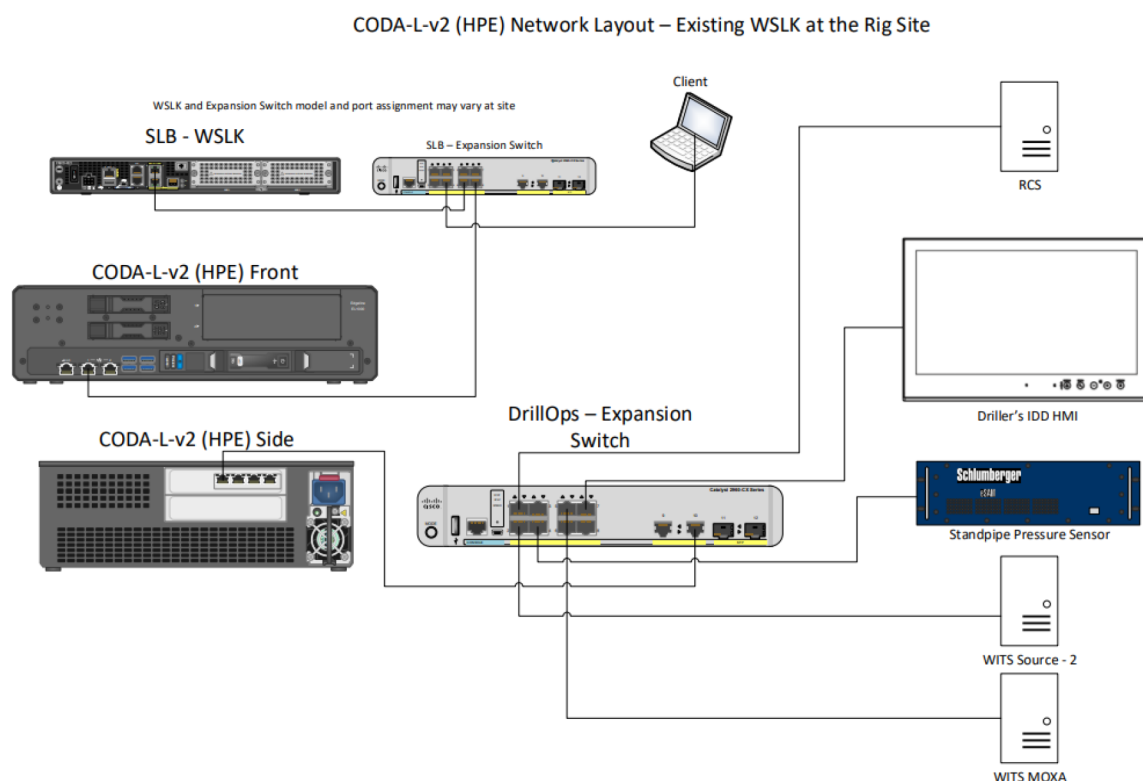


Figure 5—Generic SLB DAS rig layout.

End user hands on training starts right after the successful I&C. Digital Drilling Engineers train drillers, Well-Site Leaders, electricians and Directional Drillers on how to utilize and interact with the system. This on-rig hands on training is the longest process of the project implementation. It could take a time of drilling two gas wells for some drillers to be fully adapted to interact and trust the multi-workflow autonomous system. Once drillers are confident to utilize the system, DAS engineer moves to town to provide remote support to end users on the rig site.

Monitoring

Following a successful execution phase, the monitoring phase starts. During this phase, the DAS engineer visits rigs from time to time to perform competency assessments for drillers and other end-users and to share new system features for major system updates. In addition, the DAS engineer would continuously update SWI/SOPs, capture and report lessons learned and they would share the feedback taken from end-users to the DAS development teams to enable system enhancements. They share performance evaluation on a daily basis with the clients and other stakeholders.

Closure

During this last phase of the project, all end users are adapted to the multi-workflow autonomous system. The client and stakeholders start to capture the value the autonomous system brings to drilling operation. Implementation system documents including SOP/SWI and performance evaluation study cases are shared with the client. The ownership to utilize the system is then moved from the deployment team to the client upon the completion of the system implementation.

Deployment Challenges. Introducing the DAS to actively-drilling land rigs involves complex challenges. Some of those challenges are common between all rigs such as performing the system commissioning on parallel to ongoing drilling operation. The DAS deployment team should be aware of ongoing operation and understand all the desired moves the DAS plans to take and the logic behind it. This is extremely important

as to intercept the system at the right time if there is a bug or undesired autonomous action taken by the DAS. The DAS team is responsible for ensuring that nobody is in a red zone during the commissioning stage. Any missed bug during in town pre-test could lead to few hours of rig NPT or a failure of the implementation at the planned time. On the other hand, there are rig specific deployment challenges as we have three different OEM RCS and rig types on this project:

- I. First implementation of the system was on two Cameron rigs. This was the first deployment of the system, which is called a field test deployment. Our deployment team decided to take the ownership to field test this technology and to report all system bugs and user enhancements to the engineering and development teams until the system became mature.
- II. HH rigs with a non-integrated Top Drive system. The Top Drive system has an NOV RCS with different control interfaces. In order to deploy the DAS on those rigs, HH OEM should integrate the Top Drive system into HH RCS then HH OEM engineers would replicate all NOV movements and features on their RCS. Then, NOV engineers run a function test and certify the top drive operational safety. Failing to perfectly replicate the OEM top drive might lead to hours or days of NPT and potential safety concerns.
- III. NOV rigs with fully integrated equipment to RCS. The complexity of this deployment comes from implementing additional drilling automation solutions to perform partial processes of off-bottom and connection procedures. Both systems should synchronize with each other and have smooth transition of authority to avoid any conflict on requests to the RCS' auto driller. Therefore, additional training required to be given in town and on rigs for end users to understand how both systems work and interact together. Since this was the first worldwide deployment of its type, DAS team works closely with engineering and developers to capture bugs, recommendations and lesson learned.

Description of Hardware / Software

This section provides a straightforward explanation of each hardware component required for deploying the DAS on drilling rigs. For each component, a brief functional overview is provided, along with references to its technical capabilities. More detailed specifications can be found in the official system manuals and deployment guides.

CODA / HPE

The CODA or HPE server is the central computing unit that hosts the DAS. It is a virtualized environment composed of multiple Virtual Machines (VMs), each handling a specific function such as deployment orchestration, data storage, firewall security, and other automation-related services. This high-frequency data acquisition allows the automation engine to dynamically adjust its models in real time. To enable proper automation, the CODA server ingests critical surface and downhole data signals, including:

- Block position
- Total hookload
- Rotary RPM and torque
- Standpipe pressure and flow rate
- Weight on bit (WOB) and rate of penetration (ROP)
- Bit depth and hole depth
- Measurement While Drilling (MWD) sensor data, such as lateral, axial, and torsional vibrations
- (Optional) Differential pressure

Well-Site LAN Kit (WSLK)

The WSLK is responsible for enabling network connectivity at the rig site. It establishes a local area network and ensures stable internet access for both system operation and remote monitoring or support. Without the WSLK, remote visualization and diagnostics of the automation system would not be possible.

Uninterruptible Power Supply (UPS)

To safeguard system integrity during power fluctuations, a UPS is installed to provide backup power and voltage regulation. This is critical for protecting sensitive hardware such as the CODA server, IDD, and network switches. Its inclusion ensures continuous operation and prevents data loss or hardware damage during unexpected electrical disruptions. The UPS shields against:

- Voltage spikes
- Brownouts (temporary voltage drops)
- Fluctuations in power supply
- Complete outages

Interactive Driller Display (IDD)

The IDD is the driller's primary interface with the DAS. Installed directly inside the driller's cabin, it allows for real-time visualization, interaction, and manual override if needed. Installing the IDD involves a dedicated deployment process, including physical mounting, operating system image loading, and commissioning with the HPE server to ensure proper communication and control.

Network Switches

Switches are deployed to manage data flow between the CODA, IDD, and other connected devices. Depending on the complexity and control requirements of the site, the team may use either managed or unmanaged switches:

- Managed switches offer advanced features such as traffic routing, Quality of Service (QoS) settings, port security, and remote diagnostics. These are ideal for larger deployments where high network reliability and segmentation are required.
- Unmanaged switches provide simple plug-and-play functionality without the need for configuration. These are suitable for straightforward installations where basic connectivity is sufficient.

The DAS's Response to Downhole Operational Challenges

A critical aspect of autonomous drilling lies in the system's ability to respond intelligently to the changing downhole environment. Variations in formation strength, the onset of drilling dysfunctions, or dynamic lithological transitions require immediate and precise corrective actions to maintain drilling efficiency and equipment integrity. The DAS addresses this need through a series of closed-loop mitigation workflows that continuously monitor real-time surface and downhole data. When specific thresholds are exceeded, the system initiates targeted mitigation strategies designed to stabilize conditions and maintain progress toward the drilling objective. The following sections outline how the DAS responds to common downhole challenges using intelligent mitigation protocols embedded within its real-time control engine.

Stick-Slip Mitigation

When torsional instability, commonly referred to as stick-slip, is detected, the DAS initiates a structured mitigation sequence based on predefined system thresholds. The mitigation workflow step numbers are detailed below and highlighted graphically in Figure 6:

1. If stick-slip severity exceeds the configured risk threshold for a specified duration, the torsional mitigation routine is triggered.
2. The DAS begins the response by reducing Weight-on-Bit (WOB) and increasing rotary speed (RPM) incrementally.
3. RPM may optionally be temporarily increased beyond standard limits to suppress the torsional oscillations and re-establish smooth rotation.
4. Once the stick-slip severity drops below the allowable threshold, the system concludes the mitigation phase.
5. Optionally, a post-mitigation routine can be engaged, in which the DAS gradually reverts the WOB and RPM changes to the settings just prior to the stick-slip onset.
6. After successful mitigation, the DAS transitions back to the ROP Optimization workflow to resume normal drilling operations.

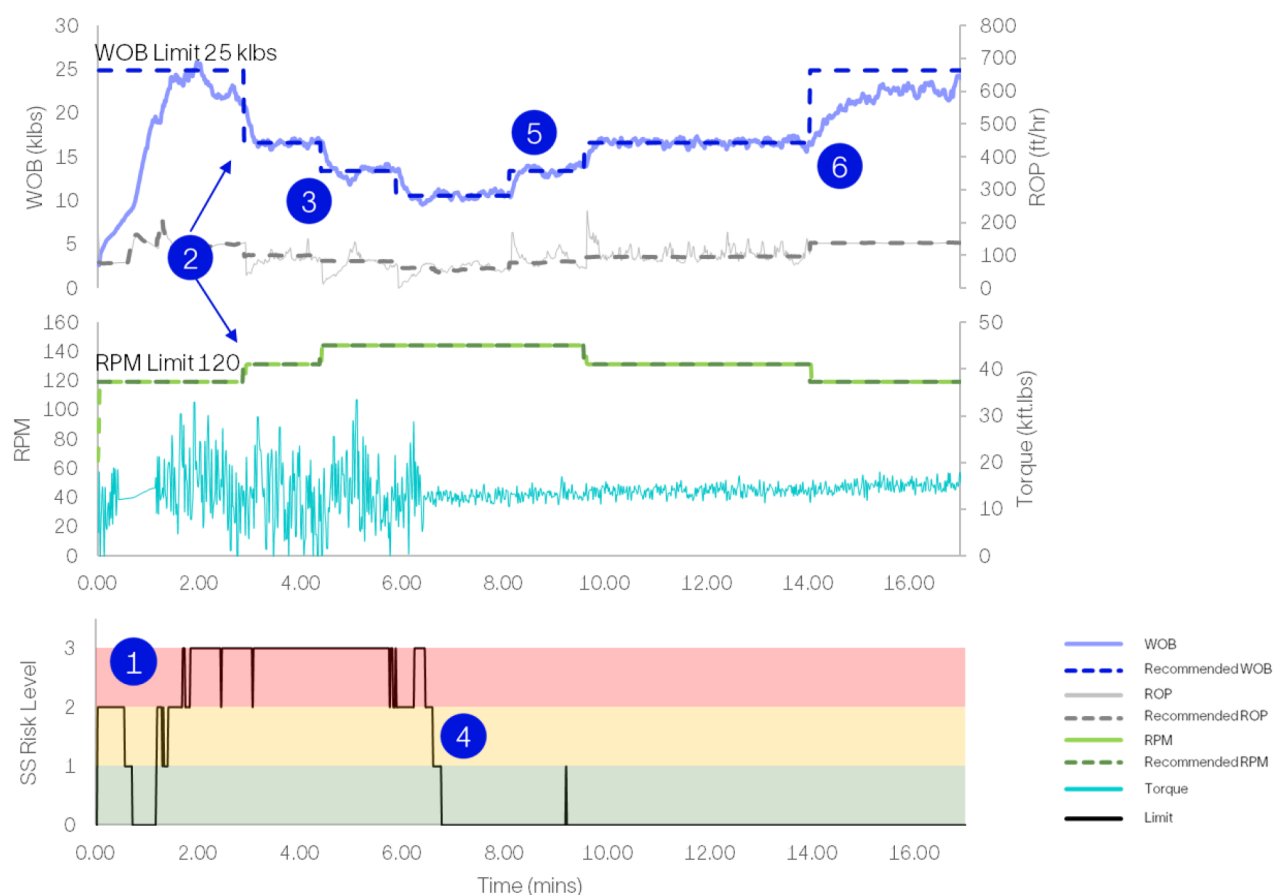


Figure 6—The stick-slip mitigation sequence detailed with surface WOB & RPM compared to SS risk level.

Lateral Vibration Mitigation

Lateral shock and vibration events can compromise tool integrity and drilling efficiency if left unaddressed. The DAS continuously monitors such conditions and reacts as follows:

1. When shock severity exceeds the configured threshold for a specified duration, the system triggers the lateral vibration mitigation routine.
2. The DAS responds by increasing WOB and decreasing RPM in defined steps.
3. Optionally, WOB can be temporarily increased beyond normal operating limits to suppress the lateral oscillation.
4. Once shock severity falls back within acceptable limits, the system ends the mitigation phase.
5. An optional post-mitigation phase may then be executed, where the DAS reverts WOB and RPM settings to values just before the event occurred.
6. Upon successful stabilization, the system returns to ROP Optimization mode.

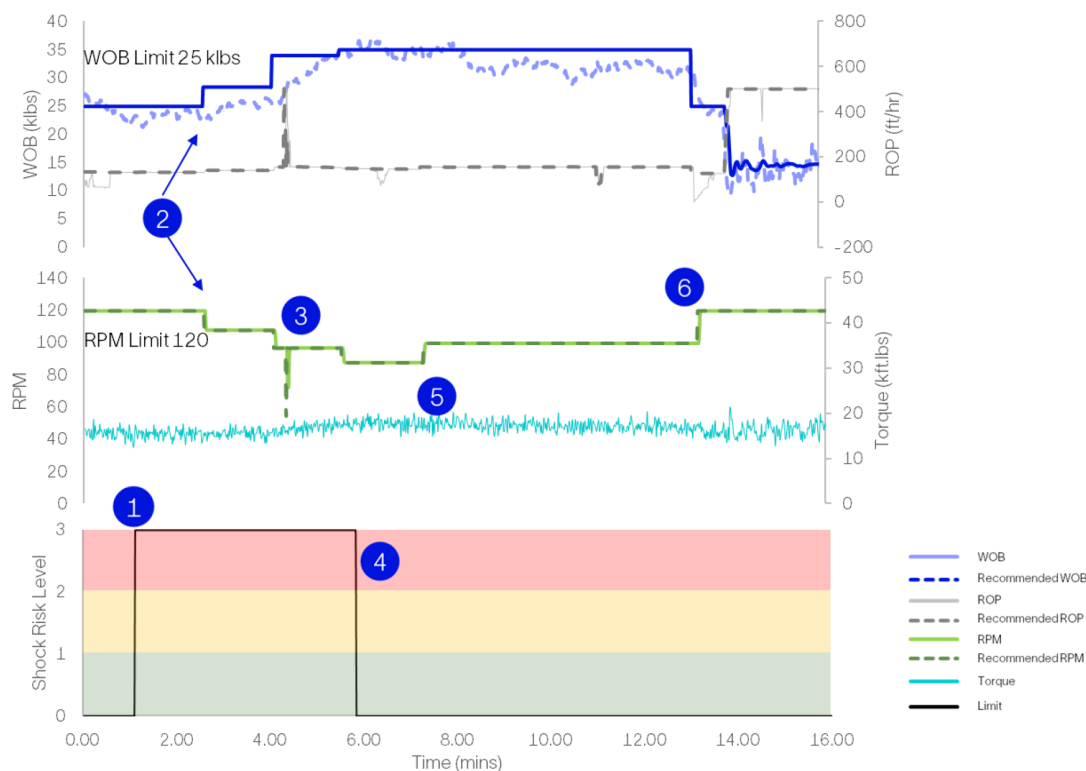


Figure 7—The lateral vibration mitigation sequence detailed with surface WOB & RPM compared to shock risk level.

Stringer Mitigation

Drilling through abrupt increases in formation strength, commonly referred to as stringers, requires a controlled response to avoid equipment overload or tool failure. The DAS identifies and addresses stringers using the following logic:

1. When a sudden change in formation strength exceeds a predefined threshold, the DAS activates the stringer mitigation routine.
2. RPM is immediately reduced to a mitigation value tailored to handle the harder interval.
3. The DAS continues drilling through the identified interval at the mitigation RPM for defined footage (e.g., several feet).
4. Once the mitigation depth is achieved, RPM is restored to the original setpoint.
5. Optionally, if a second stringer is detected upon exit from the first, the mitigation process is repeated for the new interval.
6. After completing the mitigation interval(s), the DAS resumes standard operations under the ROP Optimization module.

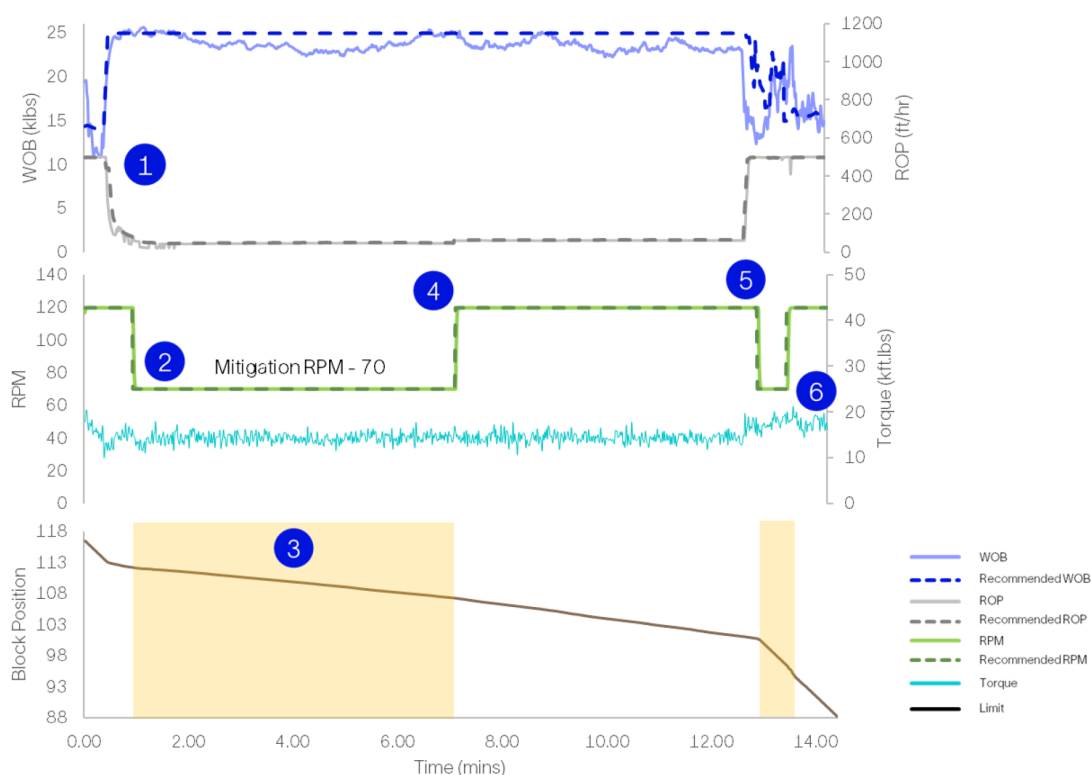


Figure 8—The stringer mitigation sequence detailed with surface WOB & RPM compared to block position.

WOB Loop vs. ROP Loop

The DAS supports two primary execution cycles to manage the drilling process based on operational priorities and downhole behavior:

- **WOB Loop:**
In this mode, WOB is maintained at a target limit, while the rate of penetration (ROP) is continuously optimized. This approach is especially effective for mitigating dysfunctions such as stick-slip or shock.
- **ROP Loop:**
Here, the system maintains a target formation penetration speed, while WOB is dynamically adjusted within permissible limits. The DAS continuously recalculates the ROP setpoint to deliver a smoother, more consistent drilling response, ideal for preventing erratic or unstable conditions.

Scalability and Operations Support Model

When implementing drilling automation, several challenges arise with one of the most critical being resistance to change in traditional workflows. An automated system demands more comprehensive planning and change management processes to ensure that system settings align with the constraints of the rig equipment and drill string.

To address this challenge, a gradual, phased approach is typically preferred by rig contractors. The first phase involves introducing the technology in "shadow mode." During this phase, the objective is to observe how the algorithm performs under real-time formation conditions and validate the outputs of the ROP optimization engine. At the end of the drilling section, a review session is conducted with the drilling supervisor and engineer to identify opportunities for optimizing drilling parameters. This review also seeks to understand the rationale behind any limiting factors. For example, weight on bit (WOB) may not have been maximized because drilling was constrained by differential pressure in certain intervals. By thoroughly

analyzing the challenges of the section and validating the ROP optimization engine's outputs, confidence is built among users, setting the stage for the next phase.

The second phase is the advisory phase. This step is particularly advantageous because it can be implemented with minimal resources and without significant modifications to the rig control system. All that is required is an edge device and a human-machine interface (HMI) to display the ROP optimization engine's recommendations to the driller in real-time. After validating the system in shadow mode, the advisory phase allows the driller to follow these recommendations. The goal is to demonstrate the value added by optimizing drilling parameters, which translates into improved ROP.

However, the advisory phase has its limitations. Over time, maintaining 100% focus on a screen becomes challenging for drillers, and the system requires frequent monitoring by engineers to ensure proper calibration and meaningful recommendations. While this phase serves as an effective proof-of-concept with short-term resource allocation, it naturally leads to the next step: full automation. Once drilling automation is implemented, the need for intensive support diminishes significantly. The system itself schedules necessary calibrations to ensure accurate recommendations, reducing the burden on personnel.

In the third phase, full drilling automation, the level of required support shifts from daily operations to performance monitoring, training new crew members, and fine-tuning system configurations from one well to the next. This transition enables remote support for multiple rigs, enhancing operational efficiency and scalability.

The successful scalability of the DAS across multiple rigs hinges on the end-user's willingness and ability to adopt the system independently. If rig personnel do not perceive the added value of the solution, there is a risk they may revert to conventional manual drilling practices, thereby undermining automation efforts. Therefore, fostering a sense of ownership among drillers and wellsite teams is critical to achieving widespread and sustained adoption.

A key factor in promoting this ownership is equipping users with the necessary skills to operate the DAS confidently without the constant presence of a dedicated automation engineer. Deploying an engineer to every rig would not only strain resources but also limit the scalability potential of the system. To address this challenge, the hyper-care training program was developed.

The hyper-care program is a structured, on-site competency development framework designed to ensure full operational readiness of rig personnel. It is built around a checklist tailored for each rig's RCS (Rig Control System) and covers three main areas:

- On-site deployment procedures
- Hands-on application of the DAS workflows
- Integration with the rig's specific RCS

Training is delivered on-site by a DAS engineer over the course of multiple drilling sections. During this period, each driller and rig manager is expected to demonstrate competence by performing the tasks outlined in the hyper-care checklist. The training culminates in a final assessment where the driller must independently connect the DAS to the RCS, configure and execute key workflows, and troubleshoot connectivity or automation-related issues in real time.

The evaluation is scored on a 100-point scale, with an 80% threshold required for passing. This ensures that only personnel who demonstrate sufficient proficiency transition to autonomous operations. By tailoring the training content to the rig's specific RCS and operational setup, the program guarantees that users gain practical experience directly aligned with their work environment. Once all rig personnel meet the competency requirements, the DAS engineer transitions to remote support for that rig. This model enables a single engineer to monitor and support multiple rigs simultaneously from a centralized location, significantly enhancing the scalability and cost-efficiency of the DAS deployment.

How Offset Data Is Used for Optimum Section Configuration

Pre-job section configuration is a critical task performed by the DAS engineer in collaboration with the well engineer / drilling engineer prior to drilling each section. During this process, the engineer inputs key parameters into the DAS, including:

- Rig-specific configuration
- Section-specific BHA details
- Operational drilling parameters
- System deployment considerations (e.g. control loops, mitigation strategies)

This section configuration acts as a launchpad for automation as it provides the driller with a ready-to-use, pre-configured workflow tailored for the upcoming section. However, it is important to note that drillers retain full operational authority. They are encouraged to adjust the configuration in real-time as needed to respond to actual downhole conditions. Since drillers are ultimately responsible for delivering the safest and highest-performing well, empowering them to make such adjustments ensures both safety and adaptability are prioritized. To support this process, the engineer develops a job record file for each rig. This file serves as a centralized reference and includes:

- The Rig survey
- I&C procedure
- The complete DAS deployment layout specific to the rig
- Visual documentation of on-site hardware integration
- Expected drilling and rig parameters for each section

After each section is drilled, a post-run analysis is conducted to evaluate the run's performance. This review identifies key successes and limitations (highlights and lowlights) and is shared with the drilling engineering team. Together, the team discusses potential modifications to the upcoming section plan to further enhance drilling performance.

In addition to individual section analysis, project-level statistical tracking is carried out. This higher-level assessment allows the DAS team to pinpoint rigs or sections that underperform relative to benchmarks. Once identified, root causes are investigated and resolved collaboratively with the well engineering team.

By documenting, reviewing, and analyzing every section drilled with the DAS, this approach fosters a culture of collaborative continuous improvement. It significantly simplifies performance optimization not only at the individual rig level, but also across entire multi-rig projects, such as campaigns involving 11 or more rigs, while minimizing time and effort from both the DAS team and drilling engineering teams.

Final Results & Conclusion

In an industry defined by complexity and high stakes, the leadership has demonstrated how innovation can transform challenges into opportunities by championing the integration of advanced technologies. We managed to redefine the execution of Lump Sum Turnkey (LSTK) projects, setting new benchmarks in efficiency, safety, and sustainability. The ability to unite cutting-edge technology with strategic vision has delivered measurable success across key domains:

- Operational Efficiency:

- Total system utilization percentage with DAS in control is above 70%
- Automation and real-time analytics have streamlined workflows which reduced execution timelines by over 4,400 hours over 122 wells.
- Total footage drilled with DAS in control is above 850,000 ft.
- Reduction in carbon emissions by over 5,000 tons of CO₂.
- Mitigation & prevention of high impact events (stuck-pipe, well control & BHA integrity) resulting in total savings of above 36,000 hours.
- Team Collaboration and Knowledge Sharing:
 - Integrated platforms have fostered seamless communication among teams, reducing delays and enhancing transparency.
 - Capturing and applying lessons learned has created a feedback loop that continuously improves project outcomes.

The combined implementation of these technologies has not only enhanced the success of individual projects but also provided a replicable framework for the oil and gas industry. By integrating automation, AI and real-time data solutions, the efforts have paved the way for a new era of digital transformation in LSTK projects. Key outcomes include:

- Scalable Solutions: Technologies that can adapt to diverse operational contexts, ensuring consistent success across global projects
- Data-Driven Decision-Making: A unified ecosystem that empowers teams to make informed decisions, reducing risk and maximizing performance
- Enhanced Client Trust: Demonstrable improvements in safety, efficiency, and sustainability have strengthened client relationships and reinforced confidence in turnkey delivery

As the industry continues to evolve, the leadership remains central to its transformation. By staying at the forefront of innovation and fostering a culture of excellence, you are not only shaping the future of LSTK projects but also contributing to the broader advancement of the oil and gas sector. This document reflects the milestones achieved and serves as a testament to the strategic vision, innovative approach and unwavering commitment to excellence.

References

- Barajas, P. E., Pastusek, P. E., Lacey, A., Sharp, M., Whitlow, Z., Glenn, L., Melton, M., Menand, S., Holder, B., Kendrick, S., and Witt-Doerring, Y. 2023. The Role of Auto Driller, Rig Control Systems and Driller Procedures in Reduction of BHA Failures. Presented at the SPE Annual Technical Conference and Exhibition, San Antonio, Texas, 16-18 October. SPE-214997-MS
- R. Covarrubias, J. Bryant, R. Israel, J. Farthing, H. Walker, C. Vahle, 2017, Development to Delivery - A Collaborative Approach to Implementing Drilling Automation, SPE/IADC Drilling Conference and Exhibition The Hague, Netherlands, 14-16 March 2017, SPE-184695-MS
- B. Goodkey, G. Hernandez, A. Nunez, M. Corona, K. Atriby, M. Rayes, R. Carvalho and C. Herrera, 2020, Drilling in the Digital Age: Harnessing Intelligent Automation to Deliver Superior Well Construction Performance in Major Middle Eastern Gas Field, Abu Dhabi International Petroleum Exhibition & Conference, Abu Dhabi, UAE, 9 - 12 November 2020, SPE-203251-MS
- A. Jourde, J. Rowatt, V. Pemmasani, B. Barbosa, JC. Yopez, P. Throw, 2020, Results from the Introduction of a Well Construction System Drilling Rig, IADC/SPE International Drilling Conference and Exhibition Galveston, Texas, 3-5 March 2020, SPE-199608-MS